

Project Controls Early Warning System V01

Project Overview

The Project Controls Early Warning System V01 is a proof-of-concept analytics solution designed to automatically identify schedule, cost, progress, and float-related issues before they escalate into major project problems.

The system transforms project controls data into actionable management alerts and recommendations that support proactive decision-making.

The project demonstrates how project performance information can be converted into an executive decision-support system using Python and analytics techniques.

Business Problem

Traditional project dashboards primarily explain what has already happened.

Project managers and executives need answers to questions such as:

- Which activities require immediate attention?
- Which activities are showing signs of deteriorating performance?
- Which activities may become critical?
- Where should management focus its efforts?
- What actions should be taken next?

The Early Warning System was developed to provide proactive alerts rather than reactive reporting.

Project Objectives

The objectives of this project are to:

- Detect schedule issues
 - Detect cost issues
 - Detect progress issues
 - Detect activities with limited schedule flexibility
 - Prioritize activities requiring management attention
 - Generate recommendations and escalation actions
 - Produce executive-level performance indicators
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Input Dataset

The project uses:

Performance_Dataset_June2026.xlsx

The dataset contains:

- Activity IDs
 - Activity codes
 - Activity names
 - Baseline durations
 - Total float
 - Planned progress
 - Actual progress
 - Progress variance
 - Budget cost
 - Planned Value (PV)
 - Earned Value (EV)
 - Actual Cost (AC)
 - Schedule Performance Index (SPI)
 - Cost Performance Index (CPI)
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Methodology

Step 1 – Schedule Monitoring

Activities are flagged when:

$SPI < 0.90$

Meaning:

The activity is significantly behind schedule.

Step 2 – Cost Monitoring

Activities are flagged when:

$CPI < 0.90$

Meaning:

The activity is spending money inefficiently.

Step 3 – Progress Monitoring

Activities are flagged when:

Progress Variance < -10%

Meaning:

Actual progress is significantly behind plan.

Step 4 – Float Monitoring

Activities are flagged when:

Float $\leq$ 5 days

Meaning:

The activity has limited schedule flexibility and may become critical.

Alert Score Logic

Each issue contributes to an overall alert score.

Issue	Score
Schedule Issue	1
Cost Issue	1
Progress Issue	1
Float Issue	2

Float receives a higher weight because activities with limited float have a greater probability of affecting project completion.

Alert Levels

Green

Healthy activity.

Yellow

Requires monitoring.

Orange

High concern requiring management attention.

Red

Executive escalation required.

Executive Recommendations

Green

Continue normal monitoring.

Yellow

Increase monitoring frequency and investigate performance trends.

Orange

Develop a recovery plan and review schedule and cost exposure.

Red

Executive escalation required. Immediate recovery actions and management review needed.

Project Workflow

Performance Dataset ↓ Issue Detection ↓ Alert Scoring ↓ Alert Classification ↓ Management Recommendations ↓ Executive KPIs ↓ Decision Support Report

Outputs Generated

The system produces:

- Issue counts
- Alert scores
- Alert levels
- Executive recommendations
- Top escalation activities
- Executive KPI dashboard
- Exportable Excel report

Technologies Used

- Python
- Pandas
- NumPy
- Google Colab
- Excel
- Project Controls concepts
- Earned Value Management principles

Skills Demonstrated

Project Controls

- Schedule monitoring
- Cost monitoring
- Progress assessment
- Float analysis
- KPI development

Analytics

- Data processing
- Rule-based classification
- Alert generation
- Management reporting
- Decision-support analytics

Construction Management

- Proactive issue identification
 - Performance management
 - Escalation planning
 - Executive reporting
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Limitations

This is a Version 1 proof of concept.

The model does not include:

- Resource loading analysis
 - Portfolio-level analysis
 - Machine learning prediction
 - Probabilistic forecasting
 - Real-time Primavera integration
 - Multi-project dependencies
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Future Improvements (V02)

Potential enhancements include:

- Machine learning risk prediction
 - Portfolio control tower integration
 - Automated email alerts
 - Interactive dashboard development
 - Trend analysis
 - Recovery scenario simulations
 - Real-time Primavera integration
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Conclusion

The Project Controls Early Warning System V01 demonstrates how project controls data can be transformed into a proactive decision-support system that helps management identify issues early, prioritize activities, and take corrective actions before project performance deteriorates.

The project integrates project controls concepts, analytics, and management reporting into a single proof-of-concept solution for executive decision support.